AHMED KHIAL

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PROFESSIONAL SUMMARY

Cybersecurity student with hands-on experience in building secure systems, threat detection, and vulnerability analysis. Skilled in developing security tools and applying machine learning techniques to enhance cybersecurity solutions, including phishing and anomaly detection systems. Experienced with Docker-based environments and scalable security architectures. Passionate about solving real-world security challenges and improving system resilience.

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EDUCATION

Bachelor of Science in Computer and Data Science (Cybersecurity Major)

Alexandria National University | Expected Graduation: 2026

Relevant Coursework: Network Security, Cryptography, Cloud Computing, Information Security Management, Risk Management, Data Integrity and Authentication, Machine Learning, Data Mining, Data Structures and Algorithms, Object-Oriented Programming, Internet of Things (IoT)

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PROJECTS

SafeSurf – Secure Phishing Detection Platform (Graduation Project)

- Developed a browser extension for real-time phishing and malicious URL detection
- Implemented rule-based techniques including URL structure analysis, entropy calculation, suspicious keyword detection, and typosquatting detection
- Built a FastAPI backend to analyze URLs and emails and return structured risk assessments (risk score, classification, reasons)
- Integrated a machine learning model to improve detection accuracy and reduce false positives
- Designed a modular multi-layer detection system combining rule-based, machine learning, and extensible security layers
- Enhancing the system with WHOIS-based domain analysis to detect newly registered domains
- Integrating threat intelligence and reputation-based checks for improved detection accuracy
- Planning a Docker-based sandbox layer for dynamic analysis and behavior inspection

Technologies: React (Vite), JavaScript, HTML, CSS, Python, FastAPI, Scikit-learn, Pandas, NumPy, Docker, Git

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Intrusion Detection System (IDS)

- Developed a machine learning-based system to detect malicious network activity
- Applied multiple algorithms including Naive Bayes, Decision Trees, Random Forests, and Neural Networks
- Evaluated model performance and optimized detection accuracy

Technologies: Python, Scikit-learn

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DNS Anomaly Detection System

- Built a system to analyze DNS traffic and detect abnormal behavior
- Identified suspicious patterns indicating potential attacks such as DNS tunneling

Technologies: Python, Pandas

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Face Recognition System

- Implemented a face recognition model using PCA and LDA with KNN
- Focused on feature extraction and dimensionality reduction

Technologies: Python, Scikit-learn

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Connect 4 AI Agent

- Developed an AI agent using the Minimax algorithm for optimal decision making

Technologies: Python

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8-Puzzle Solver

- Implemented informed and uninformed search algorithms
- Compared efficiency and performance between different approaches

Technologies: Python

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Smart Traffic and Sensor-Based System

- Designed a basic traffic control system using sensors
- Simulated automated decision-making logic

Technologies: Arduino, Embedded Systems

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TECHNICAL SKILLS

- Programming: Python, JavaScript, Java, PHP
- Web & Backend: HTML, CSS, React, FastAPI
- Cybersecurity: Threat Detection, Phishing Analysis, Vulnerability Assessment, Network Security
- DevOps & Systems: Docker, Linux, VMware, Git, MySQL
- Data & Machine Learning: Scikit-learn, Pandas, NumPy

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CERTIFICATIONS & TRAINING

- Google Cybersecurity Professional Certificate
- AWS Academy Cloud Foundations
- AWS Academy Cloud Architecture
- Cybersecurity Boot Camp (NTI)
- Embedded Systems (Arduino)
- IT Careers Program (STRA-city)
- Inoveegypt Winter Program (TIEC, ITIDA)

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ADDITIONAL INFORMATION

- Strong analytical and problem-solving skills
- Fast learner with ability to adapt to new technologies
- Interested in cybersecurity engineering, threat detection, and security systems
- Actively developing real-world security tools and scalable detection systems